

Don't Stop Me Now: Visualizing Disruptions in Railroad Networks

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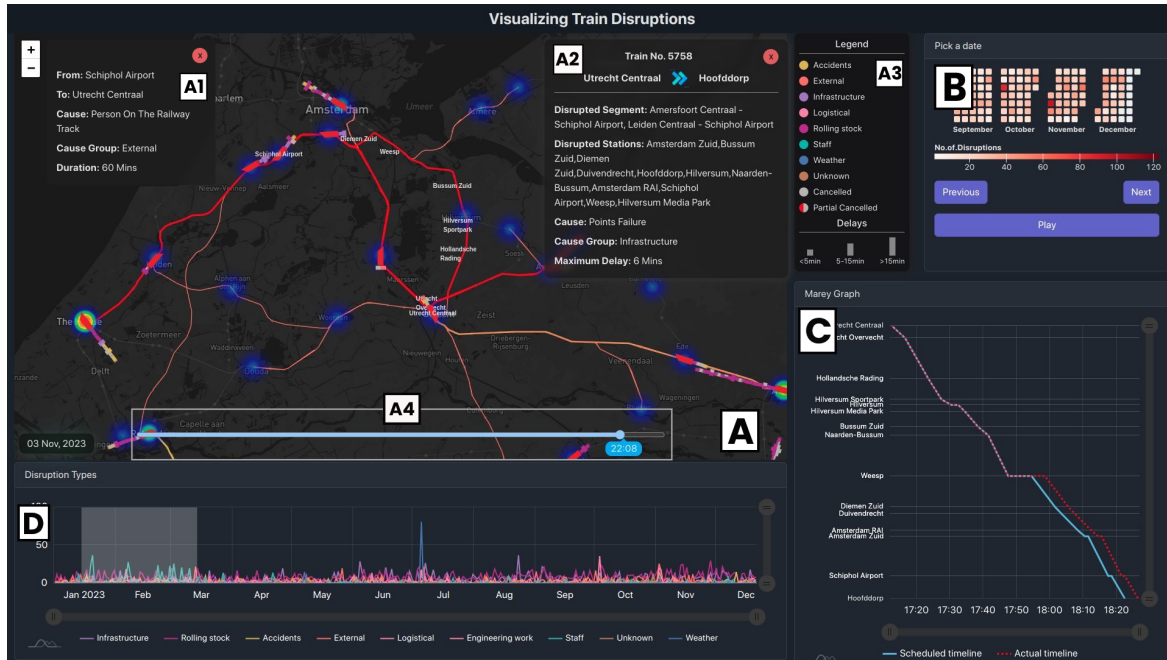


Figure 1: Overview of our approach: (A) Map information cards for disrupted segments (A1), services (A2), legend (A3), and time slider (A4); (B) Calendar heatmap showing the disruptions per day; (C) Marey graph displaying delay accumulation for selected services; (D) Multi-line chart for different disruptions over time.

ABSTRACT

Railroad networks are complex systems where a single disruption can have cumulative effects, impacting services' planned schedules and degrading the network's performance. To support railroad planners in identifying these disruptions, we propose a Visual Analytics approach with multiple coordinated views: a map, calendar heatmap, Marey graph, and a multi-line chart. Our proposed approach enables the exploration and analysis of the disruptions' spatial and temporal patterns and delay accumulation to identify vulnerable segments in the railroad network. We assess the effectiveness of our approach through an expert interview highlighting how the accumulation of delays and disruptions is lucidly communicated and provides valuable insights into their spread across the network. We discuss the outcomes of the expert interview alongside the limitations we identified and how we resolve these. Finally, we illustrate directions for future work, including online data to assist railroad planners with real-time monitoring for proactive decision-making and improved operations.

Index Terms: Spatio-temporal Data, Networks

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1 INTRODUCTION

The railroad network forms the backbone of the Dutch transportation infrastructure, enabling seamless connections across the country and the mobility of millions. According to data from Statistics Netherlands [5], in 2022, trains accounted for approximately 15.8 billion passenger kilometers. However, disruptions have become increasingly significant, with delays and uncertainties growing over time. On average, **11** disruptions occur daily, highlighting the pressing need to address the challenges faced by railroad systems [28]. Train disruptions can have a variety of causes, e.g., technical failures, weather conditions, or operational inefficiencies, and their ripple effects can be far-reaching. Understanding the frequency and patterns within disruption data is essential for designing strategies to improve the network's resiliency [13]. Visualizing various disruptions over time and space can highlight trends, peaks of instability, and high-risk locations, contributing to more effective operational planning [11]. While numerous approaches exist [30, 40, 26, 19] to improve schedules and optimize railroad operations, they often assume disruptions as predefined input parameters for simulations or optimization models. However, there remains limited attention to understanding how, when, and where disruptions emerge and accumulate. Including the spatial dimension is crucial to identify geographically recurring issues such as delays consistently originating from specific stations or regions, help reveal infrastructural bottlenecks, environmental vulnerabili-

ties, or interdependencies tied to locations [25, 29]. Our work addresses this gap by focusing on the visual exploration of disruptions to reason about their origins and accumulation across the network.

We introduce a Visual Analytics (VA) approach to explore and analyze train disruption data. Our main contributions are:

- A VA approach for exploring and analyzing railroad disruption data, supporting pattern discovery and insights (see Section 3.1);
- A visualization integrating the spatio-temporal facets of disruptions' delay accumulation in railroad networks (see Section 3);
- Expert interview with real-world data, demonstrating our approach's effectiveness and discussing limitations (see Section 4).

2 RELATED WORK

Research on transportation network vulnerability is essential for identifying bottlenecks (i.e., graph connectivity) and optimization problems (i.e., shortest path and cut edges or vertices) that impact overall network stability [14, 17]. Identifying the critical nodes and edges aids in understanding the vulnerabilities of transportation systems [8, 20]. A common underlying factor contributing to such vulnerabilities is the presence of disruptions. Disruptions affect the network and create vulnerabilities that propagate and amplify delays [14]. Extensive research has been conducted on these topics across various domains, including transport connectivity [23], scheduling optimization [30, 40, 26, 19], trip planning assistance [38], disruption management [16, 11], and delay prediction [33, 4, 13, 40]. Dekker et al. [7] investigate cascading effects of delays, highlighting the intricate interdependencies in railroad operations. This phenomenon is reinforced by [6, 4], demonstrating how disturbances ripple through railroad schedules and trigger chain reactions that disrupt the entire network. While previous research has addressed various aspects of delay management and visualization, there remains a need for approaches that integrate geospatial aspects of disruptions to better support reasoning about how, where, and why vulnerabilities emerge and propagate in transportation networks. Andrienko et al. [3] emphasize the importance of such spatio-temporal integration, yet current tools remain largely temporal and underexplored in incorporating spatial reasoning.

Over the years, various visualization techniques such as metro maps [24, 37], geospatial networks [31, 3], and glyph-based approaches [36, 27] have been applied to understand complex transportation data from both the public's and expert's perspectives. While geospatial network visualization has been explored extensively in fields like cartography and urban analytics [31], its integration into disruption management remains limited. The Marey graph (see Figure 1-C and Figure 4) is a well-designed spatio-temporal visualization that effectively represents train routes, schedules, and delays. Its effectiveness has inspired numerous studies and visualizations [39, 26, 19, 21], highlighting its adaptability and relevance for analyzing complex dynamic data. However, while powerful for visualizing delay progression, the Marey graph obscures the network's spatial aspects, making it difficult to reason about spatial relationships and disruption hotspots. Slingsby and Hyde [32] introduced "ChartTables", a zoomable level-of-detail approach to interpret probabilistic model outputs for reactionary train delays. Their approach focuses on the temporal propagation of delays using stochastic simulations but also lacks geospatial information. Fang et al. [11] apply a vulnerability analysis framework to China's high-speed rail system, using geospatial maps to highlight disruption-prone regions. While they acknowledge the spatial distribution of impacts, their model remains analytical and lacks visual support for the spatio-temporal data to explore disruption causality. Overall, existing visualizations typically treat disruptions as effects on network topology, altering routes, connections, and schedules. Yet, there's a compelling need to view disruptions geospatially to identify how they impact the railway infrastructure [25, 29] and reason about their causes, locations, and patterns. This gap paves the way

for visual approaches that not only reflect the system's response to disruption but also help explain its underlying spatial behavior.

3 VISUAL DESIGN

In this section, we introduce our proposed VA approach and the design considerations. To frame this investigation, we formulated some common concerns found across literature concerning railway infrastructure [13, 7, 9]. These concerns focus on the need to reason about disruption dynamics from both spatial and temporal perspectives, recognizing where disruptions occur, how they accumulate, and when patterns intensify. Within this context, we formulated the following research questions:

Q1: Which parts of railroad networks are more susceptible to disruptions, and how do they impact the network?

Q2: How do disruptions spread through the affected routes and services, and how do they impact overall system performance?

Q3: How do disruptions evolve, and what spatiotemporal patterns can be observed in their occurrence and accumulation of delays?

As a case study, we analyze train disruptions in the Netherlands that we obtain from Rijden de Treinen [28]. While the immediate context is centered on the Netherlands, our approach generalizes to other spatio-temporal train disruption data. The dataset includes a hierarchical categorization of disruptions, organized by cause groups (e.g., weather, staff, rolling stock) and their corresponding specific causes. Additionally, Rijden de Treinen provides train service details, including *planned* (i.e., scheduled) and *actual* (i.e., real) arrival and departure times. Representing disruptions in a way that captures their spatial and temporal context presents challenges, particularly in conveying delay accumulation and the complexity of real-world railroad operations. This is crucial for identifying critical stations and routes [14, 20] that represent vulnerabilities (**Q1**). Our goal was to design a novel visualization that would not only highlight where and when disruptions occur but also help in reasoning about their broader implications on the network's vulnerability (**Q1**, **Q2**, **Q3**).

3.1 Visualization Approach

Figure 1 shows our VA approach providing an interactive exploration environment with multiple coordinated views. The source code is available [online](#). The design offers a comprehensive overview of the train disruption data, while interactive features enable users to dive into details. The approach consists of: (A) A geospatial map (see Figure 1-A) displaying disruptions across the Netherlands, offering spatial insights into disrupted areas; (B) A calendar heatmap (see Figure 1-B) that highlights disruption frequency over time, guiding users to identify trends; (C) A Marey graph (see Figure 1-C) visualizing delay accumulation along routes and assisting users in tracking delay progression; and (D) A multi-line chart (see Figure 1-D) illustrating the trends of disruption types under different time granularities. By linking these views, the application supports users in gaining deeper insights into the spatiotemporal aspects of disruptions and their accumulation effects, thereby addressing our key research questions (**Q1**, **Q2**, **Q3**).

Disruptions in Spatial Context - Map We represent the railroad network on a geographic map to preserve spatial context [18], which facilitates pattern recognition and supports key analytical tasks such as identifying disruption hotspots, tracing their spread, and reasoning about spatial vulnerabilities [15, 31]. Alternatives like abstract node-link diagrams were initially explored for simplicity and abstraction, but we observed that these often led to visual clutter in dense areas or under-represent actual rail segments, which reduced their practical utility for planners.

The main view contains a geospatial map overlaid with an explicit network layout of the railroad system (see Figure 1-A). For an accurate representation of the railroad infrastructure, we use OpenStreetMap [1] to provide a realistic depiction of the train network.

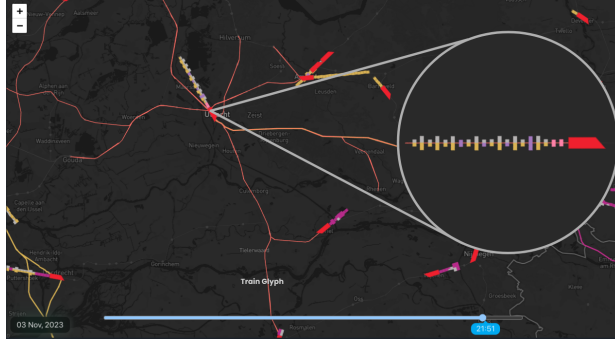


Figure 2: Map and a close-up of the train glyph. The glyph shows a train icon (in red) oriented in the direction of its destination. Bars are appended and indicate the individual disrupted services.

At the core, the stations are represented as nodes, and the edges correspond to the railroad tracks. This layout supports intuitive reasoning about system-wide vulnerabilities (Q1, Q2). When a user selects a date from the calendar heatmap (see Figure 1-B), the disruptions and the affected services for that day are loaded and displayed on the map. A time slider enables users to interactively view disruptions unfold minute by minute across the network for a single day (Q3). As time progresses, disruptions become visible and are color-coded by disruption type and encoded by line thickness (duration), helping users distinguish brief versus prolonged disruptions at a glance (Q1). Clicking on any disrupted segment reveals a detailed information card (see Figure 1-A1) containing contextual metadata: origin and destination stations, cause group, specific cause, and the duration of the disruption.

Disruptions in Spatial Context - Glyph To encode disruptions' service-level details, we adopt a design inspired by Wunderlich et al [38] (Figure 2, consisting of two components: the train body and its attached compartments (see Figure 2). This choice was driven by the need to maintain semantic familiarity with transportation icons and provides a scalable metaphor for future multi-modal systems.

We explored various designs for encoding disruption types (see Figure 3). At first, we used icons (see Figure 3-(a)) to represent different disruption causes (e.g., weather, technical faults), combined with color for intensity. However, we found that at small scales, the icons lose their readability, and users may misinterpret or miss them entirely. We, therefore, shifted to a color-based approach utilizing the pre-attentive processing of color for encoding disruption types, improving legibility without adding cognitive overload (see Figure 3-(b)). Another aspect we iterated on was how to represent multiple disrupted services. We investigated stacking compartments inside and outside the train body (see Figure 3-(c) and Figure 3-(d)), arranging them according to temporal occurrences of the disruptions (see Figure 3-(e) and Figure 3-(f)). These variants are designed with the mindset of the train body representing a single train service, which introduces challenges such as limited space, especially in regions with high traffic. We settled on an external bar layout (see Figure 2) where compartments represent individual disrupted services. Each train body summarizes multiple services arriving at that station. The train body is colored red to indicate the status of services (i.e., disrupted) and will adapt to encode real-time status in future applications. The color of the compartments indicates the type of disruption, while their height reflects the delay duration. The glyphs are placed at the destination stations, reinforcing the relationship between disruptions and where they ultimately impact passengers, offering planners insights into which parts of the network accumulate delays (Q1). Canceled and partially canceled services are represented in gray and gray-red, respectively, to



Figure 3: Design Considerations. Visualizing disruption types: (a) icon-based vs. (b) color-based encoding for better readability. Delay bar designs: (c-f) show disruption types within a single train service, using stacked compartments (c, d) and temporal ordering (e, f).

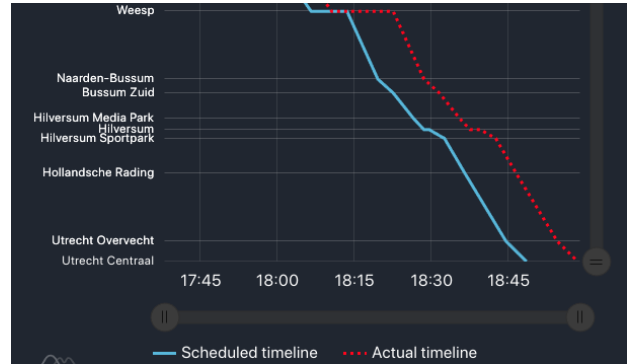


Figure 4: Marey Graph: Comparison of scheduled (solid blue line) versus actual timeline (dashed red line) for a selected service.

convey missing or incomplete data clearly. A legend (Figure 1-A3) clarifies the encoding for consistent interpretation.

Time-to-time encoding (i.e., animation) is a common way of representing dynamic data [2, 31] and therefore, as time progresses, the compartments are dynamically added based on the train's journey. This allows users to see the real-time progression of disruptions and how and when they affect different segments of the railroad network (Q1, Q2, Q3). While animation has known limitations for tasks requiring precise comparison [12], we apply it primarily to support the sequential unfolding and temporal overview of disruptions. When users click on a compartment, it opens an information card (see Figure 1-A2) to display key details (i.e., the service's origin and destination, the disrupted segment it passes through, the cause of the disruption, and the duration of the delay). Along with this information card, the entire route taken by that service is highlighted in red. To draw attention, the disrupted segment will continue flashing until the user closes the information card. This helps users in correlating services with the disrupted segments (Q1, Q2).

Disruptions in Temporal Context. Figure 1-B shows the calendar heatmap to highlight peaks of disruptions (colored in increasing red hues) over a year. Calendar heatmaps [22, 34] are widely used to identify patterns and trends over time. They are commonly featured in various applications [21, 35, 19], making them a fitting choice for this context. This visualization enables users to quickly pinpoint days with severe disruptions and periods of instability (Q3). The choice of a red color evokes a sense of urgency and highlights the negative impact of disruptions.

(C) Delay Accumulation of Disruptions. We selected a Marey Graph [10] (see Figure 4-C), to plot the selected service's planned versus actual schedules, illustrating the time taken to complete their journeys. The divergence between the lines shows the severity and progression of delays. The planned and actual schedules are rep-

resented as [blue solid](#) and [dashed red](#) lines, respectively. When users select a disrupted service (i.e., compartment) from the map view, the corresponding details are shown in an information card (see Figure 1-A2) and the relevant trajectory appears in the Marey Graph. This integration provides a detailed view of delay accumulation along the routes of train services, showing how delays accumulate (Q2). The red color and dotted line style convey a sense of delay and uncertainty in the plotted schedule.

Temporal Distribution of Disruption Types. We visualize the occurrence of disruption types over time using a multi-line chart, showing a line per disruption type (see Figure 1-D). This temporal overview allows users to spot patterns such as peaks, anomalies, or recurring trends that may correlate with meteorological conditions or seasonal operational behaviors (Q3). Each disruption type is color-coded to ensure identifiability, with red and green tones intentionally avoided. To support both high-level and fine-grained exploration, users can toggle time granularity (daily, monthly, or yearly [2]). Users can select a date of interest, e.g., when a disruption type peaks, and use this insight to explore related details in the map. For example, as can be seen in Figure 1-D, during the months of January and February, the visualization reveals a notable increase in staff-related disruptions, suggesting potential seasonal staffing challenges. Selecting such peaks enables users to explore corresponding spatial patterns on the map, revealing where and how disruptions accumulate along the routes (Q1, Q2). Users can also filter disruption types of interest, limiting the map view to show only relevant disruptions. This selective focus reduces visual clutter and helps direct attention toward specific disruption.

4 EVALUATION & FINDINGS

To evaluate the effectiveness and usability, we conducted an expert interview with a railroad network expert. The expert has developed custom tools from the open railway data source [1] giving them a strong familiarity with railroad representations. This background enabled them to use our VA approach from an informed perspective, simulating the role of a railroad planner during the evaluation. The expert was not involved in the design of our VA solution.

Expert Interview. We used the service and disruption datasets from the year 2023. The session was conducted as a semi-structured expert interview and consisted of three stages: (i) an overview of the target users' responsibilities and tasks; (ii) an introduction to our system, visual encoding, and interactions; and (iii) an open-ended exploratory scenario conducted by the participant. In total, the interview lasted approximately an hour. All protocol details and a transcript of the interview are provided in the [supplementary material](#). We encouraged the participant to think aloud—verbalizing their thought process and asking questions if they came up. Concluding the session, we conducted a structured discussion. The participant answered targeted questions related to our tasks (Q1, Q2, Q3) and provided open-ended feedback on the approach's strengths, limitations, and areas for improvement.

Discussion. We summarize the most relevant points of discussion with the expert in the following:

Q1: The expert interview suggests that the participant was able to identify disrupted segments across the network and distinguish disruption types using the color encoding. The participant noted, *"Somehow yes, you can see that many trains got delayed – I still have problems pinpointing the location of the cause."* The participant found it difficult to correlate disrupted segments with train services of interest due to the lack of a direct visual connection between disruptions and the affected services. Because of this, the participant preferred to have the train glyphs placed at the disrupted segment rather than at the destination, as this would reduce reliance on their mental map, which increases cognitive load. While the visualization partially supports Q1, adding a direct mapping between disrupted segments and their impacted train services would

improve situational awareness for railroad planners, enabling more targeted intervention.

Q2: The participant found the VA approach effective in illustrating how delays accumulate across the railroad network. The dynamic addition of compartments in the train glyphs was particularly well received. As the participant noted: *"How delays accumulate, especially with the glyphs where they get appended – that's interesting."* This quote highlights how the train glyph intuitively conveys the buildup of disruptions along the routes. Additionally, the Marey graph further enhanced the analysis by offering a temporal view of delay accumulation, allowing users to track delay progression. Together, these visual views enabled a comprehensive understanding of both localized and network-wide delays.

Q3: The participant found it easy to identify peak disruption periods using the calendar heatmap and the disruption type chart. *"Yes sure, with the disruption type graph. It's easy to find the days that have the most disruptions."* During the session, the participant was particularly drawn to a prominent spike on July 5th (see Figure 1-D), which was attributed to weather-related disruptions, sparking further exploration in the map. This highlights how the coordinated views can guide users in connecting temporal peaks with spatial patterns and underlying causes. They noted the date picker as a useful feature for quickly accessing days of interest and suggested adding a global filter for consistent analysis by disruption type.

Improvements According to the outcome of the expert interview, we implemented the following improvements. We expanded the map and Marey graph and unified all views into a single frame, addressing previous issues faced by the participant when interacting with the layout. We added a heatmap overview [39] to the map view and a blinking animation to highlight disrupted segments when a service of interest is selected, improving the correlation between disruptions and services. Disruption types are filterable through the multi-line chart to reduce visual clutter of the map view. We remark that the visualization we describe in Section 3 already includes these changes. Screenshots of both the evaluated and earlier versions are provided in the [supplementary material](#).

5 CONCLUSION & OUTLOOK

We presented a VA approach for identifying and analyzing railroad disruptions and assessed the effectiveness through an expert interview. The evaluation highlights strengths in visualizing delay accumulation and disruption types, while also revealing challenges in correlating disruptions with specific services and routes. We outline the limitations we identified and our outlook on how these can be addressed, as well as directions for future work. Challenges from high station density lead to visual clutter on the map. To mitigate, we plan to explore abstract or schematic representations that allow flexibility to distort geospatial data (i.e., metro maps [24, 37]), along with more compact and adaptive glyph designs to improve clarity and scalability in dense network regions. While our approach provides valuable insights into the disruption dynamics, several areas for future work remain. Integrating online data will be a key step to support real-time monitoring and enable more responsive actions by planners. Incorporating predictive models to forecast system recovery (i.e., when disruptions stabilize) could support proactive decision-making, with our interface serving as a front-end to such models by overlaying predicted delays or highlighting potential disruption cascades based on historical patterns. The study also revealed an evolution in disruption types, highlighting an interesting aspect for further exploration.

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